

MR Credit: Building Life-Cycle Impact Reduction

1–5 Points

This credit applies to

- BD+C: New Construction
- BD+C: Core & Shell
- BD+C: Schools
- BD+C: Retail
- BD+C: Data Centers
- BD+C: Warehouses & Distribution Centers
- BD+C: Hospitality
- BD+C: Healthcare

INTENT

To encourage adaptive reuse and optimize the environmental performance of products and materials.

REQUIREMENTS

Demonstrate reduced environmental effects during initial project decision-making by reusing existing building resources or demonstrating a reduction in materials use through life-cycle assessment. Achieve one or more of the following options for a maximum of 5 points.

Option 1. Building and Material Reuse (1–5 points)

Maintain the existing building structure, envelope, and interior nonstructural elements. Reused or salvaged materials from off site that are incorporated into the building can also contribute to the credit calculations. However, reuse materials contributing toward this credit may not contribute toward MR Credit Material Disclosure and Optimization – Sourcing of Raw Materials.

Historic, abandoned, or blighted buildings: Portions of buildings deemed structurally unsound or hazardous can be excluded from the credit calculations.

Path 1: Maintain Existing Structural Elements – Walls, Floors, Roofs, and Envelope (1–5 points)

Maintain the existing building structure (including floor and roof decking) and envelope (the exterior skin and framing, excluding window assemblies and nonstructural roofing materials). Calculate reuse of the existing project area according to Table 1.

Table 1. Path 1 Points for reuse of existing building structural elements

Percent of existing walls, floors and roof reuse by project area	Points BD+C
15%	1
30%	2

45%	3
60%	4
75%	5

Path 2: Maintain Interior Nonstructural Elements (1 point)

Use existing interior nonstructural elements (e.g. interior walls, doors, floor coverings and ceiling systems) for at least 30% of the entire completed building, including additions.

Option 2. Whole-Building Life-Cycle Assessment (1–4 points)

For new construction (buildings or portions of buildings), conduct a cradle-to-grave life-cycle assessment of the project's structure and enclosure and select one or more of the following paths to earn up to 4 points:

- **Path 1:** Conduct a life cycle assessment of the project's structure and enclosure (1 point).
- **Path 2:** Demonstrate a minimum 5% reduction in at least three of six impact categories, one of which must be global warming potential (2 points).
- **Path 3:** Demonstrate a minimum 10% reduction in at least three impact categories, one of which must be global warming potential (3 points).
- **Path 4:** Meet Path 3 requirements AND incorporate reuse/salvage materials; demonstrate at least 20% reduction in GWP and 10% in two additional impact categories (4 points).

For Paths 2, 3, and 4, no impact category may increase by more than 5% compared with the baseline building. The service life of the baseline and proposed buildings must be the same and at least 60 years. Data sets must be compliant with ISO 14044.

Impact categories:

- Global warming potential (greenhouse gases), in kg CO₂e
- Depletion of the stratospheric ozone layer, in kg CFC-11e
- Acidification of land and water sources, in moles H⁺ or kg SO₂e
- Eutrophication, in kg nitrogen eq or kg phosphate eq
- Formation of tropospheric ozone, in kg NO_x, kg O₃ eq, or kg ethene
- Depletion of nonrenewable energy resources, in MJ (CML) / depletion of fossil fuels (TRACI)

GUIDANCE – Further Explanation

Beta Update

Updates in v4.1 include a greater focus on reducing embodied carbon of building structures and enclosures. Former LEED v4 Options 1 and 2 (historic building reuse and abandoned/blighted building) have been combined into Option 1 for v4.1. The former Credit Option 3 (Building and Material Reuse) is replaced with a consolidated Option 1 including two pathways restoring the LEED v2009 credit pathways. Credit Option 2 (formerly Option 4), Whole Building Life Cycle Assessment, now has an entry pathway that rewards the effort to conduct a WBLCA without requiring specific impact reductions.

Calculations – Path 1 Sample

Table 1. Sample Building Structure and Envelope Reuse Calculation for Path 1

Structure/Envelope Element	Existing Area (sf)	Reused Area (sf)	Percentage Reused (%)
Foundation/slab on Grade	11,520	11,520	100.0%
2nd floor Deck	11,520	10,000	86.8%

1st floor interior structural Walls	240	240	100.0%
2nd floor interior structural Walls	136	136	100.0%
Roof Deck	11,520	11,520	100.0%
South Exterior Wall (excl. windows)	8,235	8,235	100.0%
East Exterior Wall (excl. windows)	6,535	6,535	100.0%
West Exterior Wall (excl. windows)	6,535	5,820	89.1%
Off-site salvage - brick facade for north exterior wall (excl. windows)	8,235	5,500	66.8%
Off-site salvage - structural beams for new stairwell	650	650	100.0%
Total	65,126	60,156	92.4%

Equation 3. Percentage of existing building reuse – maintain existing structural elements:

$$\text{Existing building reuse} = (\text{Total Reused On-site} + \text{Total Reused Off-site}) / \text{Existing Building Area} \times 100$$

Calculations – Path 2 Sample

Table 2. Sample Interior Nonstructural Element Reuse Calculation for Path 2

Interior Non-Structural Element	Total Area* (sf)	Existing/Reused (sf)	Percentage Reused (%)
Gypsum Board Wall Partitions – full Height	5,400	2,500	46.3%
Gypsum Board Wall Partitions – Partial Height	650	650	100.0%
Carpeting	15,000	0	0.0%
Resilient flooring	350	350	100.0%
Ceramic tile	150	150	100.0%
Suspended ceiling systems	10,400	6,500	62.5%
Gypsum Board ceilings	350	350	100.0%
Interior Doors (Wood)	525	420	80.0%
Interior Windows / sidelights	56	56	100.0%
Offsite salvage: interior Doors (Metal)	86	86	100.0%
Offsite salvage: interior casework / cabinetry	235	235	100.0%
Totals	33,202	11,297	34.0%

*Note: the total area calculation includes both new and existing/reused materials.

Equation 4. Percentage of existing building reuse – maintain interior nonstructural elements:

$$\text{Interior nonstructural reuse} = (\text{Area of Retained Elements} + \text{Area of Reused Off-site Items}) / \text{Total Area of Interior Nonstructural Elements} \times 100$$

GUIDANCE – Option 2 Whole Building LCA Tool Guidance

The system boundary must include modules A-D as defined in ISO 21930 and EN 15804. Required modules include:

- Product stage: A1-A3
- Construction process: at least A4
- Use Stage: at least one module from B1-B5
- End of life stage: at least one module from C1-C4

LCA software must have ISO-14044-compliant data sets and conform to ISO 21931-2017 and/or EN 15978:2011.

EXEMPLARY PERFORMANCE

- Option 1 Path 1: Reuse 90% of the building
- Option 2: Achieve Path 4 and show 40% reduction in GWP
- Achieve more than 5 points from a combination of Option 1 and Option 2

REQUIRED DOCUMENTATION

- Path 1: Structural and nonstructural reused elements table and calculations
- Path 2: Interior nonstructural reused elements table and calculations
- Option 2: WBLCA report for structure and enclosure of building
- Option 2, Paths 2-4: WBLCA report including description of LCA assumptions, scope, analysis process, life cycle impact summary, and narrative indicating which path was pursued and how reductions were achieved

CONNECTION TO ONGOING PERFORMANCE

- LEED O+M MR prerequisite Waste Performance: The concept above is a collection of strategies that can help achieve MR prerequisite Waste Performance in the O+M v4.1 rating system.